Complex Analysis Homework #8

1. Find all holomorphic functions $\widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$.

If a function $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$ is holomorphic then it restricts to a holomorphic function $\mathbb{C} \to \mathbb{C}$, since $\mathbb{C}$ is open in $\hat{\mathbb{C}}$. So we first consider holomorphic functions $\mathbb{C} \to \mathbb{C}$ and then understand how they can be extended to $\hat{\mathbb{C}}$. Recall the holomorphic functions $\mathbb{C} \to \mathbb{C}$. These are exactly functions with a single power series representation

$$f(z) = c_0 + c_1 z + c_2 z^2 + \cdots,$$

with infinite radius of convergence (i.e. such that $\limsup |c_n|^{1/n} = 0$).

To extend a holomorphic function $f : \mathbb{C} \to \mathbb{C}$ to a function on $\hat{\mathbb{C}}$ we must only specify its value at ∞. To verify that it's a holomorphic extension we must then check that our extended function is holomorphic at ∞.

If $f : \mathbb{C} \to \mathbb{C}$ holomorphic is constant $f(z) = k$ then, by continuity, we must have that $f(\infty) = k$. In this case, $f : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ is holomorphic since the extended function is a constant function on $\hat{\mathbb{C}}$. On the other hand, if f is non-constant then it must be unbounded. I.e. $\lim_{|z|\to\infty} f(z)$ converges for all approaches and takes value ∞. And so, again, by continuity we must have $f(\infty) = \infty$. To make sense of this last statement we can view our extended f "in the other patch" given by the transition $z \mapsto 1/z$. In this sense, we can view $f(\infty)$ as $f(1/z)$, and we have defined this value to be 0 as $z \to 0$. We have argued how to continuously extend a non-constant holomorphic function from $\mathbb{C}$ to $\hat{\mathbb{C}}$, but we need to argue that this extension is holomorphic at $z = \infty$.

■

2. Let

$$f_n(z) = \sum_{k=0}^{n} z^k/k!.$$

Fix $R > 0$. Prove that for n large enough (depending on R), f_n has no zeros in $D(0, R)$.

First recall that the argument principle states, given a holomorphic function $f : U \to \mathbb{C}$, an open set $V \subseteq U$ with $\bar{V} \subseteq U$ and f nonvanishing on ∂V, we have

$$\int_{\partial V} \frac{f'(z)}{f(z)} dz = 2\pi i \sum_{p \in V} Mult_p(f).$$

In particular, it is sufficient to show then that there is an n large enough so that $\int_{\partial D(0,R)} (f_n)'/f_n dz = 0$.

First notice that

$$(f_n)'(z) = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^{n-1}}{(n-1)!}.$$

And so

$$\frac{f'(z)}{f(z)} = 1 - \frac{z^n}{n!} \frac{1}{f(z)}.$$

Then

$$\begin{aligned} \int_{\partial D} \frac{f'}{f} dz &= \int_{\partial D} 1 - \frac{1}{n!} \frac{z^n}{f(z)} dz \\ &= -\frac{1}{n!} \int_{\partial D} \frac{z^n}{f(z)} dz. \end{aligned}$$

By Cauchy's integral theorem, since $f(z) = 1$ is entire. And now, we can understand this remaning integral in terms of the residue theorem

$$\int_{\partial D} \frac{z^n}{f(z)} dz = 2\pi i \sum_{p \in D} res_p(z^n/f_n(z)).$$

It is only possible for us to have a non-zero residue at a point p if $f_n(p) = 0$. And so, let us consider where the zeroes of $f_n(z)$ may lie. In particular, if there exists an n such that all the zeroes of $f_n(z)$ lie outside of $D(0, R)$ then we have shown what we have set out to show.

I wasn't totally sure how to bound the magnitude of the zeroes from here. One thing I thought of doing was attempting to consider $f_n(z) = e^z - \sum_{k=n}^{\infty} z^k/k!$ and so $f_n(z) = 0$ if

and only if $e^z = \sum_{k=n}^{\infty} z^k/k!$. I could attempt to apply log to both sides and then poke at the symbols.

Here's another idea that I thought about. Suppose that for all n there existed some z_0 a zero of f_n with $|z_0| \leq R$ for all n. Since the series $\lim_{n\to\infty} f_n(z) = e^z$ converges uniformally, I wonder if this might imply that e^z has a zero in $\mathbb{C}$ (which would be a contradiction). But I'm not sure I have a theorem that says anything like this at hand.

$\blacksquare$

3. Let G be the group of Mobius transformations. Show that any $M \in G$ is conjugate in G to either $f(z) = \lambda z$, for some $\lambda \in \mathbb{C}^*$, or to $f(z) = z + 1$.

First I claim that the number of fixed points of a Mobius transformation is an invariant of its conjugacy class. Suppose a Mobius transformation is represented by $A \in PGL_2(\mathbb{C})$ and suppose it fixed a vector v. Then a conjugation BAB^{-1} for some $B \in PGL_2(\mathbb{C})$ is tantamount to a change of basis of A. In particular $(BAB^{-1})(Bv) = Bv$ and so BAB^{-1} fixes Bv — a fixed vector of A maps to a fixed vector of any of its conjugates. Moreover, this process is inverted by conjugating again $B^{-1}(BAB^{-1})B = A$. And so, since Mobius transformations are in particular bijections, the number of fixed points is a well-defined invariant of its conjugacy class.

We consider the possible number of fixed points of a Mobius transformation. By triple transitivity we know that there's a unique Mobius transformation which sends one triple of distinct points to another triple of distinct points. And so if a Mobius transformations fixes at least three points, it must be the identity. Hence, a non-identity Mobius transformation must fix either $0, 1, 2$ points. Neither of these transformations appear to fix zero points. Note, by inspecting the equation $f(z) = z$ for a general Mobius transformation, I think we can show that there's no Mobius transformation which fixes zero points.

Now consider the given functions. Note that $\lambda z = z$ is solved only by $z = 0$ if $\lambda = 1$. If $\lambda = 1$ then $f(z) = \lambda z$ is the identity map. In both cases $f(z) = \lambda z$ also fixes the point at infinity. Likewise $z + 1 = z$ is not solved for $z \in \mathbb{C}$, however, the point at infinity is fixed by this map. Hence the maps $f(z) = \lambda z$ and $f(z) = z + 1$ have two fixed points and one fixed point respectively.

We have argued that the number of fixed points is well-defined invariant of the conjugacy classes of $PGL_2(\mathbb{C})$, and that a non-identity Mobius transformation can only have $0, 1$, or 2 fixed points. But we also need to argue that there are only three conjugacy classes in $PGL_2(\mathbb{C})$. That is, a priori there could be multiple conjugacy classes of $PGL_2(\mathbb{C})$ with a given number of fixed points.

Apologies, the following will be a bit rambly. I was hoping to characterize the Mobius transformations based on the solutions to the equation $f(z) = z$. Let $f(z) = (az + b)/(cz + d)$ be a Mobius transformation. Note that $f(z) = z$ is given by the quadratic equation

$$cz^2 + (d - a)z - b = 0.$$

First suppose that $c = 0$, then this equation has two distinct solutions in z if $(d - a)^2 + 4bc \neq 0$. Otherwise it has at least one solution.

In the case where $c = 0$ our fixed point equation becomes

$$(d - a)z - b = 0.$$

If $d = a$ then our Mobius transformation is equivalent to $f(z) = (z + \tilde{b})/1$ which only has a fixed point at infinity. Likewise if $d \neq a$ then we have a fixed point $z = b/(d - a)$ but also our Mobius transformation is equivalent to $f(z) = \tilde{a}z - \tilde{b}$.

Without loss of generality, we can choose our coordinate system so that one fixed point is at ∞. For ∞ to be fixed we must have $c = 0$ I am guessing here, I suppose this needs to be shown better. Then the fixed point equation $f(z) = z$ becomes $(a - d)z = -b$. From this point I would try to explicitly construct the change of basis matrices which sends f to λz if $a \neq d$ (I suppose this is the transformation which further sends $-b/(a - d)$ to 0), and the change of basis which sends f to $f + 1$ if $a = d$. ∎

4. Let $f(z) = \frac{az+}{cz+d}$ be a Mobius transformation. Show that the number of rational maps $g : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ such that

$$g(g(g(z))) = f(z)$$

for all z is either $1, 3$, or ∞. (Hint: use previous problem)

First note that it is sufficient to find $g \in \mathbb{C}(x)$ such that $g^{\circ 3} = z + 1$ or $g^{\circ 3} = \lambda z$ for $\lambda \neq 0$. If $g^{\circ 3} = hfh^{-1} \in PGL_2(\mathbb{C})$ then notice

$$(h^{-1}gh)^{\circ 3} = h^{-1}g^{\circ 3}h = h^{-1}(hfh^{-1})h = f.$$

That is, if we find a rational g with $g^{\circ 3}$ in some conjugacy class of a mobius transformation, then we have found the g such that $g^{\circ 3} = f$ for all f in the same conjugacy class.

Consider first g as a degree 1 rational function. Note that we can represent degree 1 rational functions as matrices with the same method that we have been representing mobius transformations as matrices (except now the representation of some rational functions will have zero determinant). And so we can use matrix multiplication to compute the compositions a bit more easily.

Consider first $f(z) = \lambda z$. Consider rational functions of the form

$$[g]^3 = \begin{bmatrix} \alpha & 0 \\ 0 & 1 \end{bmatrix}^3 = \begin{bmatrix} \alpha^3 & 0 \\ 0 & 1 \end{bmatrix}.$$

And so we will have $g^3 = f$ for each solution of $\alpha^3 - \lambda = 0$ in $\mathbb{C}$. But recall that $\mathbb{C}$ is algebraically closed and so for a fixed λ we have three solutions to this equation. The discriminant to this cubic equation is $\Delta = \lambda(-27\lambda + 18)$ is not zero unless $\lambda = 18/27$. And so for most λ we have three distinct solutions to this equation. And so, we have at least three solutions to $g^3(z) = \lambda z$. could there be any other solution?

how about for the other function? For $f(z) = z + 1$ we have that

$$\begin{bmatrix} 1 & 1/3 \\ 0 & 1 \end{bmatrix}^3 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

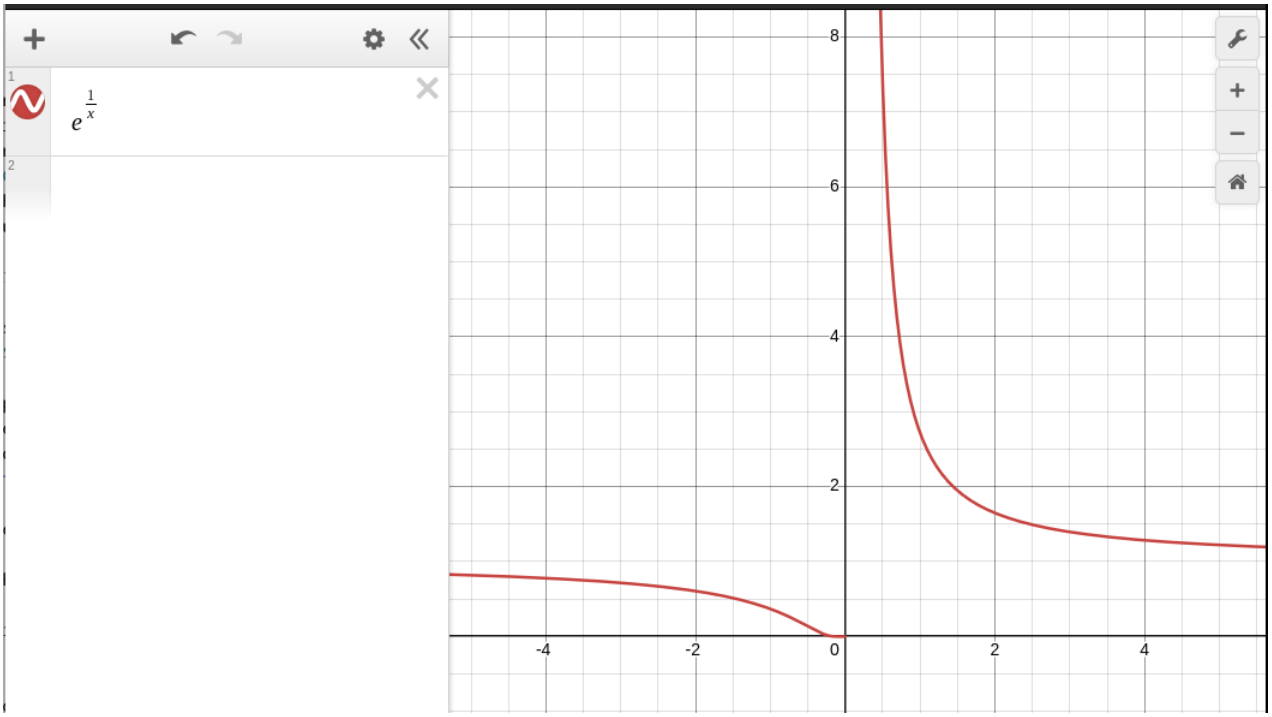
However we haven't said anything about whether any other degree 1 map might cube to give the desired map. In particular, I think it is possible for a power of a non-upper triangular matrix to become upper triangular.

Lastly, we would need to address non-degree 1 rational functions.

■

5. Construct an explicit sequence $z_n \to 0$ such that $\exp(1/z_n) \to 0$.

First consider the following plot of $Re(e^{1/z})$ from desmos.



This plot suggests that a sequence approaching 0 along the negative real axis may give us what we want. Let $z_n = -1/n \in \mathbb{R}$. We have that $z_n \to 0$. First notice that

$$\lim_{n\to\infty} 1/z_n = \lim_{n\to\infty} -n" =" -\infty$$

And we so we have

$$\lim_{n\to\infty} e^{1/z_n} = \lim_{u\to-\infty} e^u = 0.$$

■

6. Prove or disprove: if $f : \Delta \to \mathbb{C}$ is holomorphic with n zeros, then f' has at least $n-1$ zeros in Δ. Do the same, with "at least" replaced by "at most".

To begin, suppose that the zeroes of $f(z)$ are $z_1, \cdots, z_n \in \Delta$. Then, I claim that we can write $f(z) = (z - z_1) \cdots (z - z_n) g(z)$ where g is a non-vanishing holomorphic function on Δ.

Recall that, by inspecting the power series of f about a zero, say z_1, we can write $f(z) = (z - z_1) g_1(z)$ locally about z_1 for some m and where g_1 is non-vanishing on a neighbourhood of z_1 (since the zeroes are isolated). But now g_1 is a holomorphic function on Δ with zeroes at the remaning points $z_2, \cdots, z_n$. And so, about z_2 we can write $g_1(z) = (z - z_2) g_2(z)$ where g_2 is non-vanishing on a neighbourhood of z_2. Note that g_2 is also non-vanishing on a neighbourhood of z_1 because g_1 is. Now, we can write $f(z) = (z - z_1)(z - z_2) g_2(z)$. Continuing this process yields the claimed form of f. Note that this construction deals with n distinct zeroes of f in Δ, however, essentially the same construction can be used if we have "repeated zeroes" (unclear to me whether that's disallowed in the question set-up).

Now applying the product rule to compute $f'(z)$ yields

$$f'(z) = \sum_{i=1}^{n} (\Pi_{k \in ([n] \setminus i)} (z - z_i) g(z)) + \Pi_{i=1}^{n} (z - z_i) g'(z)$$

And then

$$\frac{f'(z)}{f(z)} = \sum_{i=1}^{n} \frac{1}{z - z_i} + \frac{g'(z)}{g(z)}$$

Recall that the argument principle will yield

$$\frac{1}{2\pi i} \int_{S^1} \frac{f''(z)}{f'(z)} = \sum_{p \in \Delta} Mult_p(f')$$

So long as $f'(z)$ does not vanish on S^1.

Now, if there were some relationship between f''/f' and f'/f I could rewrite the desired integral in terms of the things that we know. I think there might be something like this. Recalling that f''/f' is called the "logarithmic derivative of f'" we could notice that

$$f''/f' = \frac{d}{dz} \log(f') = \frac{d}{dz} \log(f'/f \cdot f) = \frac{d}{dz} \log(f'/f) + f'/f,$$

However, applying something like this to the desired integral is not super clear to me. On the other hand the f'/f term contains the term $g'(z)/g(z)$, the integral of which should come to 0 since g is nonvanishing on Δ.

Moreover, the question is asking "prove or disprove". And so, I could have attempted this question by inspecting particlar holomorphic functions on the unit disc, and attempted to inspect their derivatives.

■